

Aunabil Chakma

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Research Interest

LLMs, In-Context Learning, Prompt Optimization, Model Alignments

Education

University of Arizona

Ph.D. in Computer Science

Tucson, AZ

Aug 2024 – Present

Bangladesh University of Engineering and Technology (BUET)

B.S. in Computer Science and Engineering

Dhaka, Bangladesh

Feb 2015 – Apr 2019

Selected Publications

- **Aunabil Chakma**, Mihai Surdeanu, Eduardo Blanco. *Structured Semantic Information Helps Retrieve Better Examples for In-Context Learning Applied to Few-Shot Relation Extraction*. **ACL 2026 (Oral)**.
- **Aunabil Chakma**, Mihai Surdeanu, Eduardo Blanco. *Two-Stage Prompt Optimization for Few-Shot Relation Extraction: From Reasoning-Guided Search to Gradient-Guided Refinement*. **ARR May 2026 (Under Review)**.
- **Aunabil Chakma**, Aditya Chakma, Masum Hasan, Soham Khisa, Chumui Tripura, Rifat Shahriyar. *ChakmaNMT: Machine Translation for a Low-Resource and Endangered Language via Transliteration*. **arXiv**, 2025.
- **Aunabil Chakma**, Masum Hasan. *Leveraging BanglaBERT for Low Resource Sentiment Analysis of Bangla Language*. **BLP Workshop @ EMNLP2023**.

Work Experience

Research Assistant, University of Arizona

Jan 2025 – Dec 2025

- Researching narrative rewriting with prompting strategies; advised by Prof. Eduardo Blanco and Prof. Joshua Garland.

Teaching Assistant, University of Arizona

Fall 2024, Spring 2026

- Supported CSC 445 (Algorithms) and CSC 450 (Algorithms for Bioinformatics) through grading and tutoring.

Software Engineer (Associate SWE, SWE I, SWE II), Therapservices

Nov 2020 – Jul 2024

Dhaka, Bangladesh

- Built HIPAA-compliant Java SaaS modules and automated backend data/file-transfer workflows for U.S. government and private organizations serving **500,000+ individuals**.

Academic Service

- Reviewer: EACL 2026 Demo; AACL-IJCNLP 2025 Demo; NeurIPS 2025 Reliable ML Workshop.
- Ad-hoc Reviewer: ARR May 2026, ARR May 2025; EMNLP 2025 Demo.

Skills

Programming: Python, Java, C++, LaTeX

ML/AI: PyTorch, TensorFlow, Hugging Face, OpenNMT-py, spaCy

Achievements

- Received **TA of the Month** in Dec 2024 and Apr 2026.
- **3rd place** (out of 30 teams), Shared Task on Sentiment Analysis of Bangla Social Media Posts, **BLP Workshop at EMNLP2023**.
- **4th place** (out of 39 teams), BUET Intra CSE-14 Programming Contest 2016.
- Solved **1000+ programming problems**; Codeforces max rating: **1706**.